LESSON PLAN

AIRCRAFT MAINTENANCE ORDER (1)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flight testing of naval aircraft, Maintenance Test Flight, Check Test Flight and documentation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flight testing of naval aircraft, Maintenance Test Flight, Check Test Flight and documentation
5. Specific Objectives	1.1.5.1 Trainee will be able to understand modification, life categories and flight-testing effect of CG variation and wartime maintenance procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight test documentation forms and wartime maintenance checklists under instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary objectives of flight testing naval aircraft?
(b) Define the difference between a Maintenance Test Flight and a Check Test Flight.
(c) Why is Center of Gravity (CG) variation critical during flight testing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of flight testing for naval aircraft • Maintenance Test Flight procedures and scope • Check Test Flight requirements and criteria • Documentation standards for flight test records • Wartime maintenance procedures and adaptations	L & Ds	PPT, IB	15 min
(b)	• Modification life categories and definitions • Flight-testing effects of CG variation • Impact of CG shifts on aircraft stability • Procedures for verifying modification limits • Safety margins during post-modification flights	L & Ds	PPT, PDF	20 min
(c)	• Review of flight test documentation forms • Identification of wartime maintenance checklists • Verification of CG calculation sheets • Analysis of maintenance test flight reports • Correction of documentation errors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how CG variation affects the outcome of a Check Test Flight.
(b) Describe the key documentation required for a wartime maintenance procedure.
(c) Differentiate between modification life categories based on flight test results.

12. Home Assignment and Follow Up	Prepare a brief summary of the loading and CG determination procedures to be covered in LP 12, highlighting how CG variation impacts flight safety.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Loading of aircraft and determination of center of gravity, effects of variation of CG on load arm	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min